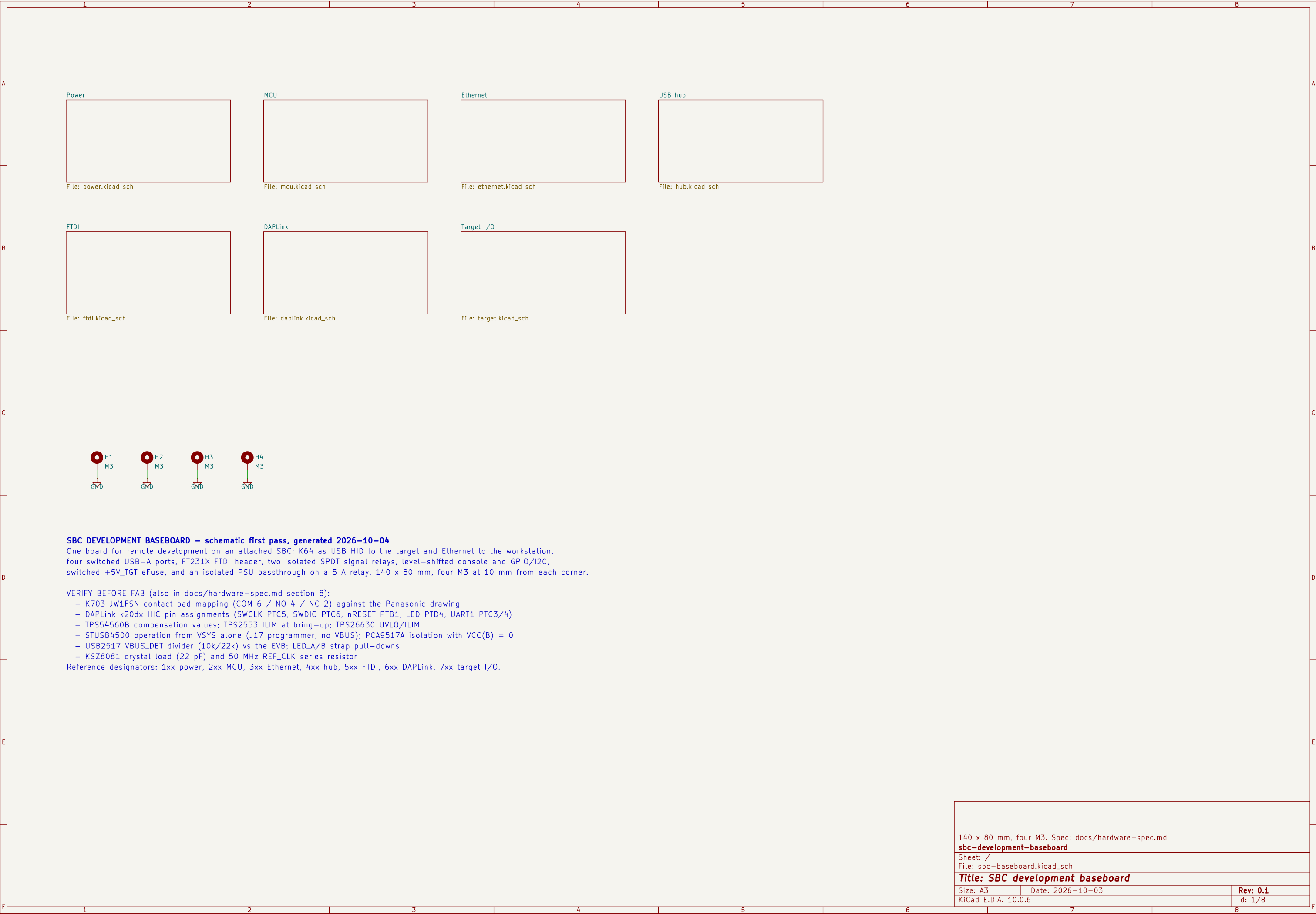
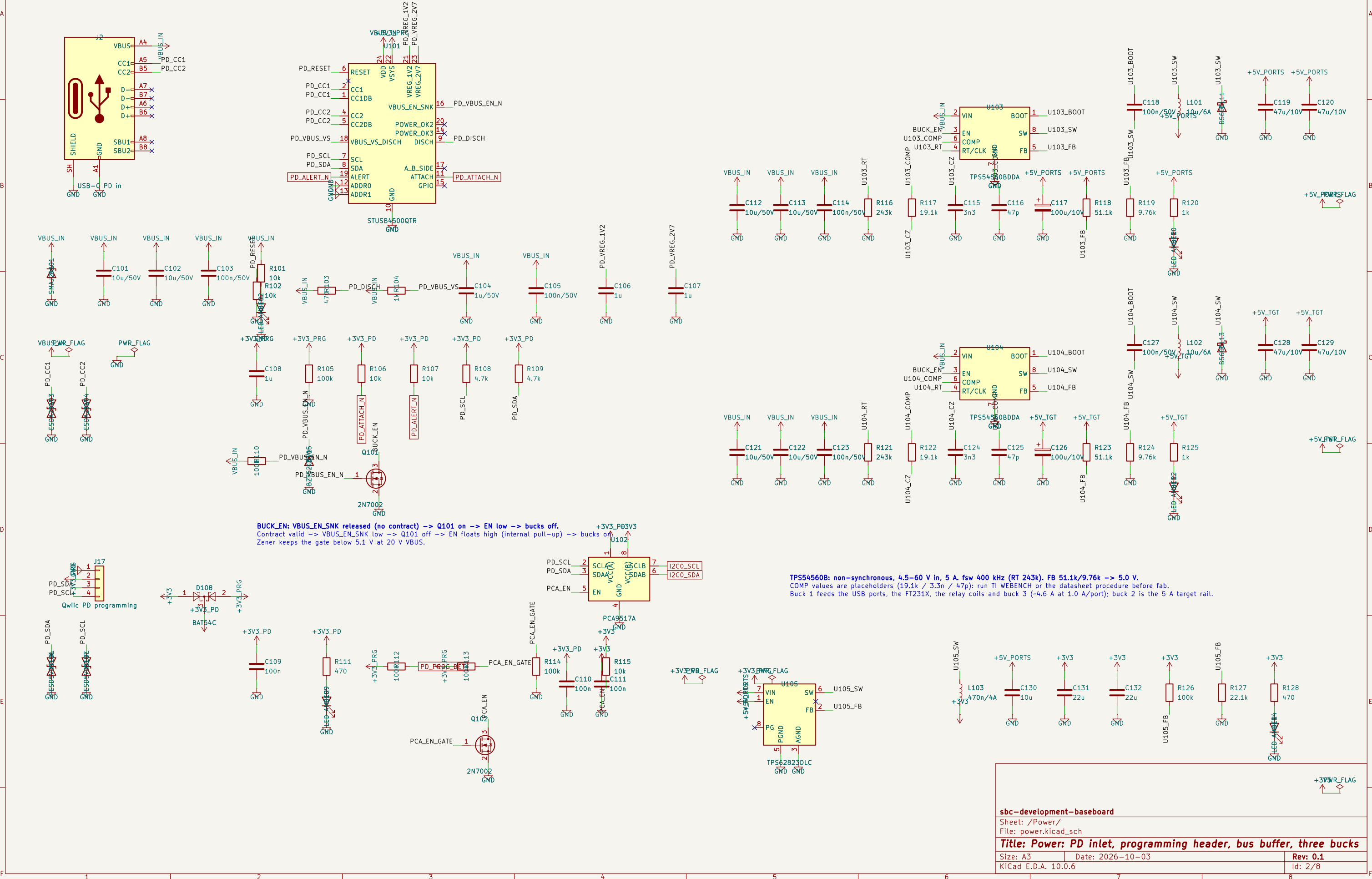




POWER

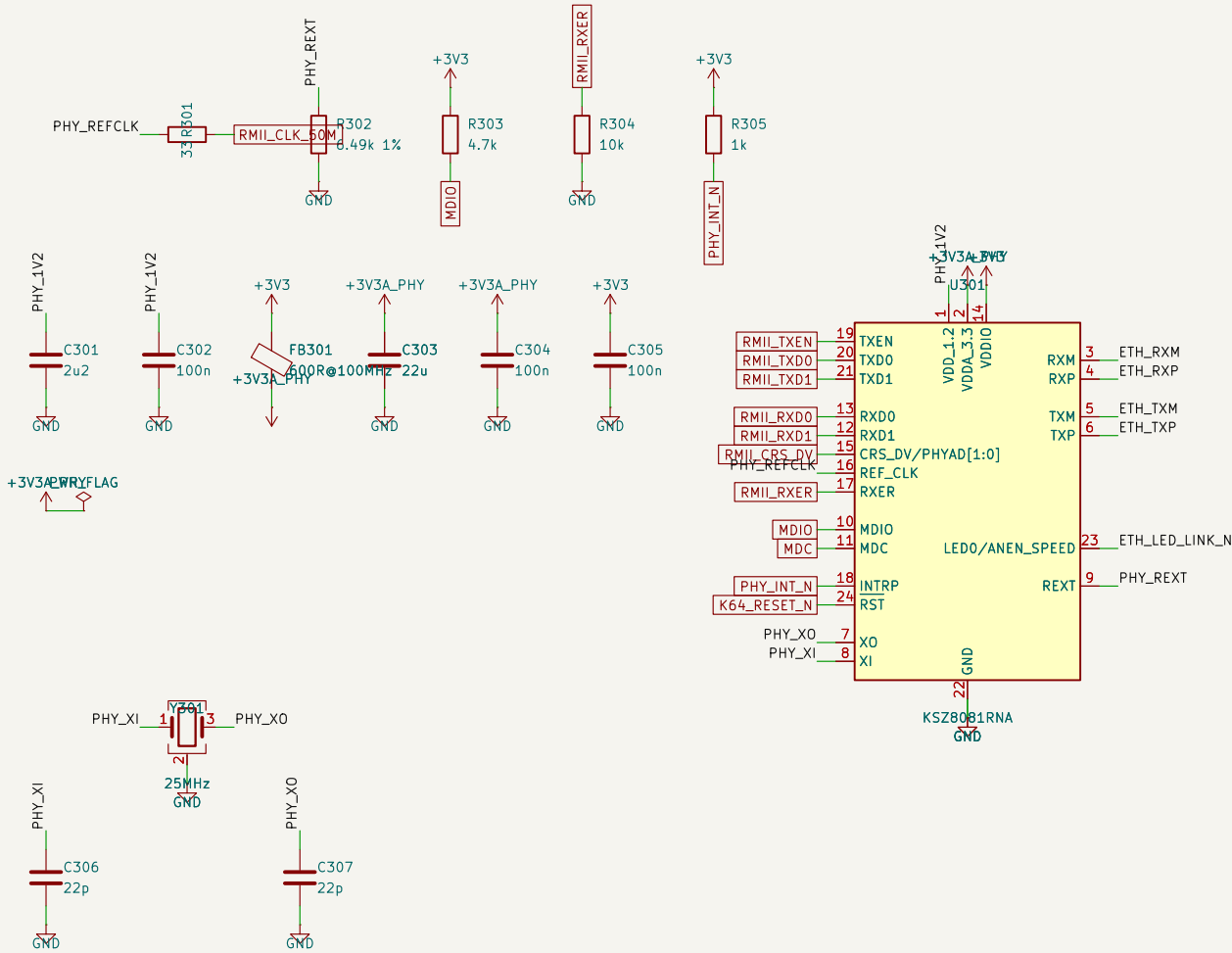
J2 is a PD sink only (no data). STUSB4500 negotiates from NVM; VBUS_EN_SNK gates the 5 V bucks, so the rails only come up once a sink contract (or plain Type-C 5 V) is valid. No series VBUS FET.
J17 (Qwiic) programs the STUSB4500 NVM and powers it through VSYS when the board is dead.
PCA9517A: A side = PD segment (+3V3_PD, diode-OR of +3V3 and the programmer), B side = K64 I2C0.
A programmer on J17 pulls PCA_EN low: the K64 is disconnected from the PD bus in hardware.



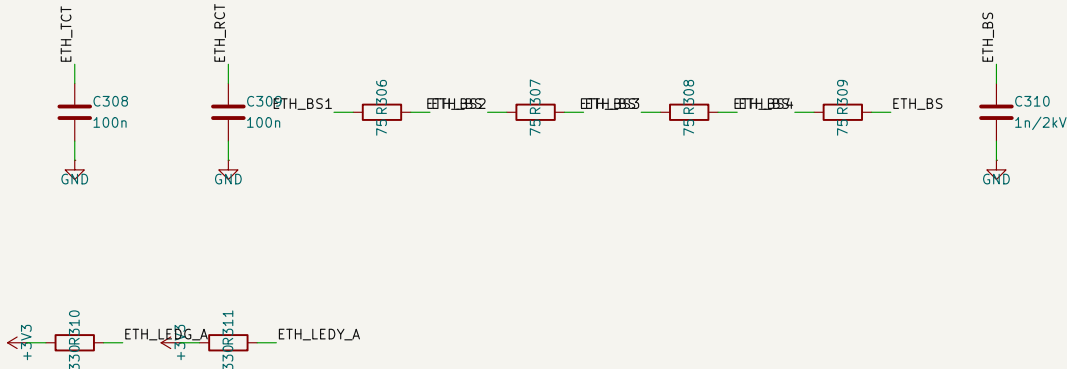
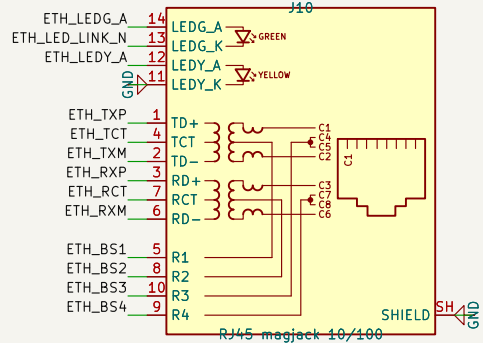
BUCK_EN: VBUS_EN_SNK released (no contract) -> Q101 on -> EN low -> bucks off.
Contract valid -> VBUS_EN_SNK low -> Q101 off -> EN floats high (internal pull-up) -> bucks on
Zener keeps the gate below 5.1 V at 20 V VBUS.

TPS54560B: non-synchronous, 4.5-60 V in, 5 A, f_{sw} 400 kHz (RT 243k). FB 51.1k/9.76k -> 5.0 V.
COMP values are placeholders (19.1k / 3.3n / 47p): run TI WEBENCH or the datasheet procedure before fab.
Buck 1 feeds the USB ports, the FT231X, the relay coils and buck 3 (-4.6 A at 1.0 A/port); buck 2 is the 5 A target rail.

ETHERNET
KSZ8081RNA in RMII 25 MHz mode (its power-up default): 25 MHz crystal on XI/X0, REF_CLK outputs 50 MHz to the K64 EXTAL0 through R301. This is the FRDM-K64F arrangement (Zephyr: microchip:interface-type = rmii-25MHz). PHY address 0 (pins 15/16 internal pull-downs; the MAC does not drive them). RXER needs the external pull-down. Transformer centre taps go to 0.1 uF each, not to a supply (datasheet 11.x). Bob Smith: 4x75R to 1 nF/2 kV.

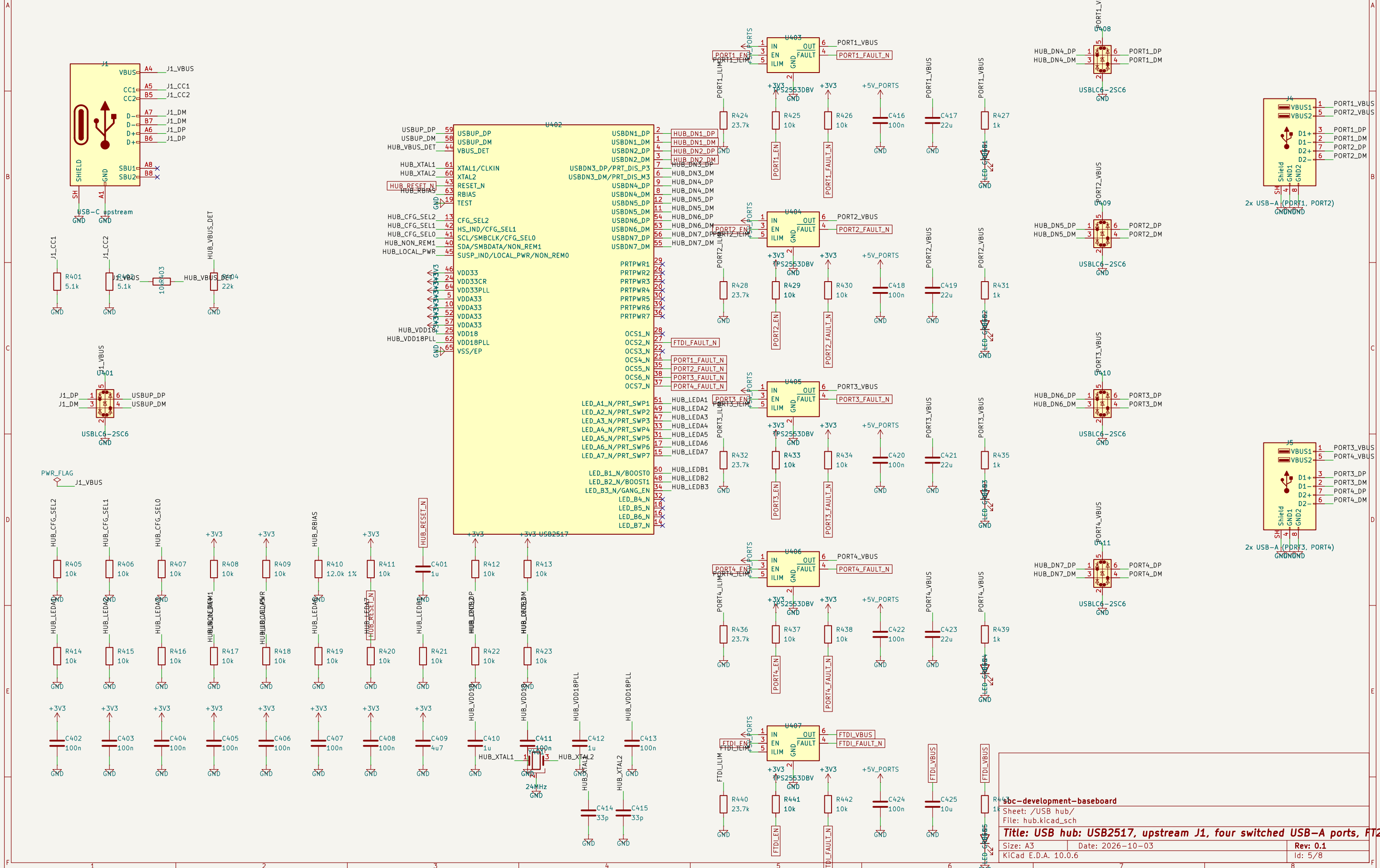


Crystal: 25 MHz +/-50 ppm, CL 16 pF typ (datasheet) -> 22 pF each, trim at bring-up.



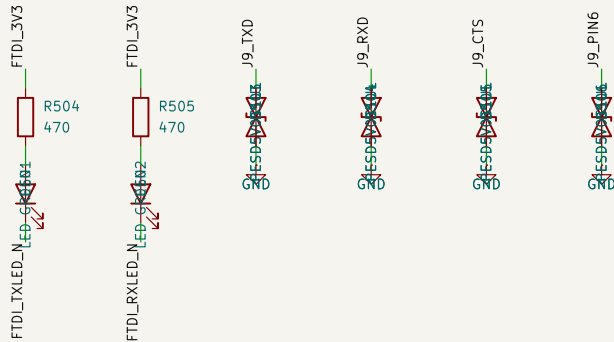
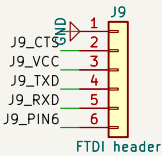
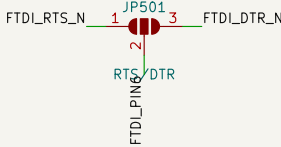
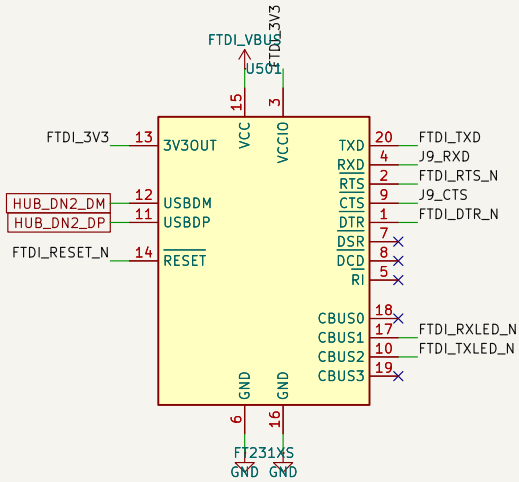
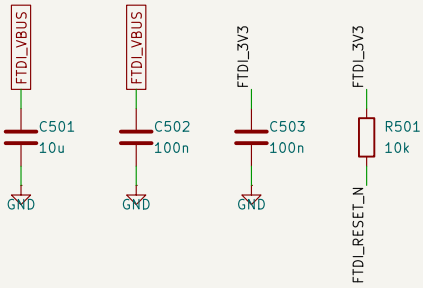
Green LED = link/activity (PHY LED0, active low). Yellow LED = PHY powered (cathode to GND).

USB HUB
USB2517 strap-configured (CFG_SEL=000), no firmware needed for the USB tree to come up.
Port map: 1 = DAPLink, 2 = FT231X, 3 = disabled (DN3 pulled up), 4..7 = USB-A PORT1..PORT4 (J4: ports 4,5; J5: ports 6,7).
NON_REM=11 (pin 40 up, pin 45 up) marks ports 1-3 non-removable and LOCAL_PWR high reports self-powered.
LED_A/B pins pulled down: PRT_SWP normal polarity, BOOST=00, GANG_EN=0 (individual over-current).
Port power is switched by TPS2553 under K64 GPIO (boot ON via pull-ups); PRT_PWR outputs unused; /FAULT feeds OCSx_N.



TPS2553 ILM 23.7k -> -1.1 A (datasheet: 20k -> 1.3 A typ, 15k -> 1.7 A). EN/FAULT pull-ups set the boot-ON state.

FTDI HEADER J9
Pinout = TTL-232R / SparkFun: 1 GND (blk) 2 CTS (brn) 3 VCC (red) 4 TXD (org) 5 RXD (yel) 6 RTS/DTR (grn).
TXD/RXD named from the FT231X side: TXD drives the target's RX. 3.3 V levels only; 1.8 V consoles use J13.
JP501: pin 6 = DTR# (bridge C-B, default) or RTS# (bridge C-A). JP502: pin 3 VCC from 3V3OUT (50 mA), open by default.
VCC comes from the TPS2553 on the hub sheet (FTDI_VBUS); FTDI CYCLE power-cycles the bridge without a replug.
CBUS defaults: CBUS1 = RXLED#, CBUS2 = TXLED# (factory MTP); change with FT_PROG if ever needed.



sbc-development-baseboard

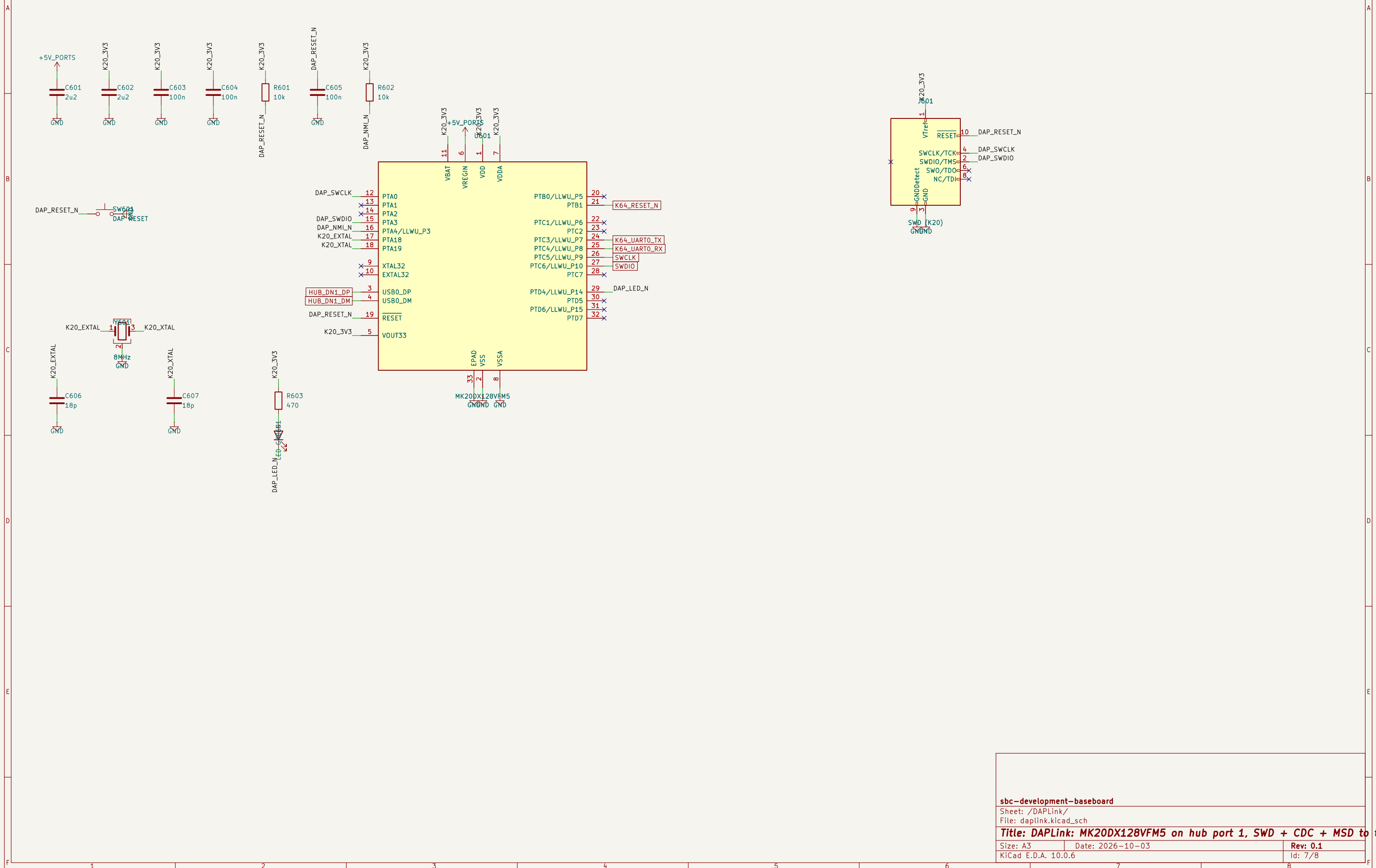
Sheet: /FTDI/
File: ftdi.kicad_sch

Title: FTDI: FT231X on hub port 2, standard 6-pin FTDI header

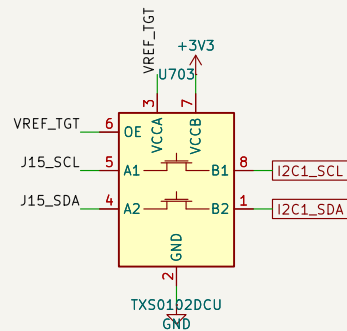
Size: A3 Date: 2026-10-03 Rev: 0.1
KiCad E.D.A. 10.0.6 Id: 6/8

DAPLINK

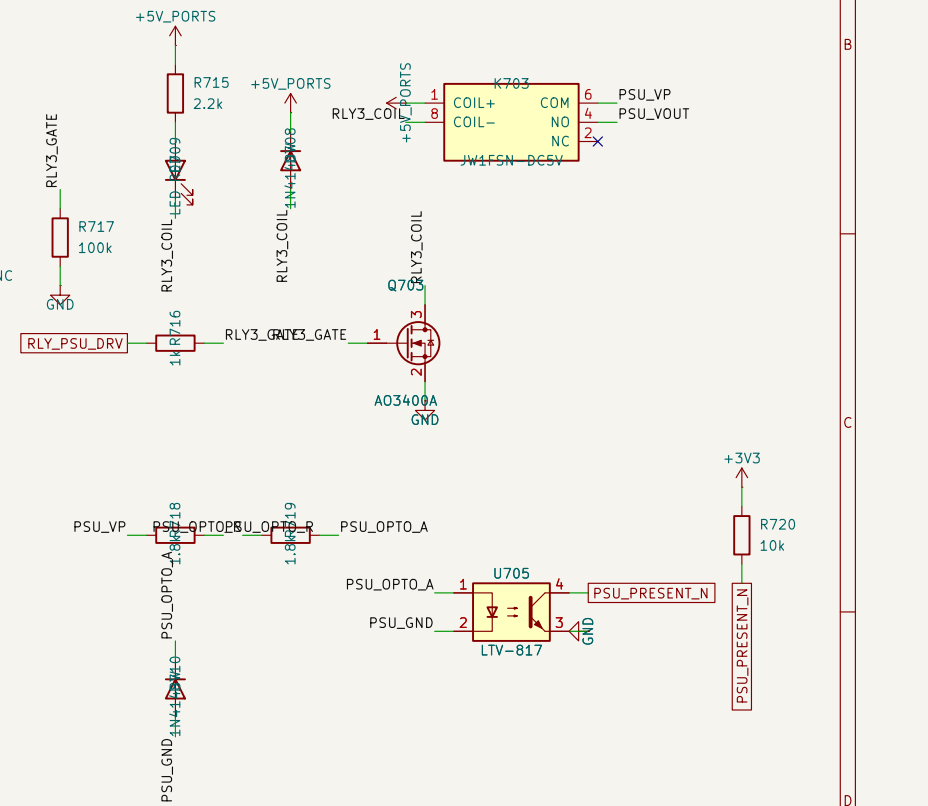
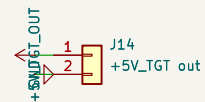
MK20DXX128VFM5 running DAPLink (k20dx HIC), powered from +5V_PORTS through its own USB regulator (VOUT33 -> VDD). Pin use follows the DAPLink k20dx HIC: PTC5 = SWCLK, PTC6 = SWDIO, PTB1 = nRESET to the K64, PTD4 = LED, UART1 PTC3/PTC4 bridges to the K64 UART0 (CDC console). Verify against DAPLink source/hic_hal/freescale/k20dx/IO_Config.h. J601 programs the K20 itself (bootloader + DAPLink). SW601 held at power-up enters DAPLink maintenance mode.



VREF_TGT comes from the target on J13.1 / J15.1 and sets the level of every translator (TXB0104/TXB0108/TXS0102). OE pins follow VREF_TGT: with the target off the translators are high-impedance. J15 GPIO are push-pull (TXB). Relays boot de-energized (gate pull-downs). K701/K702 G6K-2F-Y: pole 1 used (COM 3, NC 2, NO 4); pole 2 unused. Passthrough J18 -> K703 (NO) -> J19 on nets PSU_VP / PSU_VOUT / PSU_GND that touch no board net. Opto only. K703 JW1FSN pad map COM=6 NO=4 NC=2: VERIFY against the Panasonic terminal drawing before fab.



Pin configuration diagram for the TDA19841 in QFN32 package. The diagram shows pins 1 through 17 with their functions and connections. Pin 1 is IN (+5V_TGT), Pin 2 is B_GATE, Pin 3 is OUT (+5V_TGT), Pin 4 is PGTH, Pin 5 is IN_SYS (+5V_TGT), Pin 6 is DRV, Pin 7 is UVLO (TGT_UVLO), Pin 8 is FLT (TGT_FAULT_N), Pin 9 is PGOOD, Pin 10 is IMON (TGT_IMON), Pin 11 is OVP, Pin 12 is SHDN (TGT_EN), Pin 13 is MODE, Pin 14 is ILIM (TGT_ILIM), Pin 15 is dVdT (TGT_DVdT), Pin 16 is GND, and Pin 17 is GND. The diagram also shows connections for +5V_TGT, U704, and GND.



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